

ANALYSTLAB AFRICA  
EXPERIENCE LAB INTERNSHIP PROGRAMME  
**HEALTHCONNECT CLINIC EXPERIENCE LAB**  
**WEEK 6 — INTEGRATION & DEPENDENCY REGISTER**

*Improving Patient Appointment Attendance and Healthcare Support Using Data and AI*

| Field              | Detail                                                                            |
|--------------------|-----------------------------------------------------------------------------------|
| Project            | Improving Patient Appointment Attendance and Healthcare Support Using Data and AI |
| Programme          | AnalystLab Africa Experience Lab                                                  |
| Track              | Project Management                                                                |
| Project Phase      | Solution Development & Integration                                                |
| Reporting Period   | Week 6                                                                            |
| Register Version   | 3.0                                                                               |
| Baseline Reference | Week 5 Risk & Dependency Register — V2.0                                          |
| Register Owner     | Project Management Function                                                       |
| Review Authority   | Project Sponsor / Project Leadership                                              |

**OVERALL STATUS: AMBER — ACTIVE INTEGRATION CONTROL**

## 1. Register Purpose

The Integration & Dependency Register is the controlled record of the conditions, outputs and interfaces that must function correctly for HealthConnect workstreams to progress from independent development toward an integrated, testable solution.

It gives Project Management a single control point for determining:

- What is required;
- Who provides it, and who depends on it;
- Why and when it is required;
- What constitutes an acceptable handoff;
- Whether the dependency has moved beyond production into actual downstream use;
- What exposure exists if the dependency is not satisfied; and
- What intervention is required to protect integration, quality, schedule or readiness.

### Control principle

*A dependency is not considered satisfied merely because the originating workstream has completed its task. Satisfaction requires evidence that the required output or condition is available to the receiving activity at the level of usability and acceptance required for downstream progression.*

## 2. Scope and Control Boundary

### In scope

The register controls material dependencies involving:

*Data Analytics ↔ Data Science ↔ ML / ML Engineering ↔ Generative AI*

and interfaces between those workstreams and Project Management, integrated solution development, acceptance activities, and Week 7 testing and refinement.

### Out of scope

The register does not replace the Project Timeline, Risk Register, Issue Log, Decision Log, Workstream Integration Map, or the Week 7 Testing & Refinement Coordination Plan. Where a dependency creates or materially changes a risk, issue, decision or schedule exposure, the relevant control record should be updated and cross-referenced.

## 3. Week 6 Control Evolution

Week 5 established the baseline dependency structure. Week 6 moves control from identifying dependencies to determining whether those dependencies are actually functioning across workstream boundaries.

| Week 5 Baseline                   | Week 6 Integration Control                        |
|-----------------------------------|---------------------------------------------------|
| Dependency identified             | Dependency actively monitored                     |
| Provider identified               | Provider and receiver accountability established  |
| Output defined                    | Downstream use defined                            |
| Dependency status recorded        | Dependency state and maturity tracked             |
| Handoff anticipated               | Handoff evidenced                                 |
| Downstream impact considered      | Integration consequence assessed                  |
| Dependency monitored              | Acceptance condition established                  |
| Failure escalated where necessary | Failure linked to the appropriate project control |

*The register therefore measures not only whether an output exists, but whether it has successfully crossed the interface for which it was produced.*

## 4. Dependency Control Model

Material dependencies are assessed through the following progression:

*Identify → Define → Assign → Produce → Transfer → Accept → Integrate → Validate → Close*

| Level                 | Condition                                                                 |
|-----------------------|---------------------------------------------------------------------------|
| L0 — Identified       | Dependency exists; required output or condition has not yet been produced |
| L1 — Produced         | Provider has produced the required output                                 |
| L2 — Transferred      | Output has been formally provided to the receiving workstream             |
| L3 — Accepted for Use | Receiver confirms that the output meets the required condition for use    |
| L4 — Integrated       | Output has been incorporated into downstream work                         |

| Level              | Condition                                                            |
|--------------------|----------------------------------------------------------------------|
| L5 — Validated     | Integrated result has been assessed against applicable criteria      |
| L6 — Testing Ready | Integrated component satisfies defined conditions for Week 7 testing |

### Control distinction

*L1 does not equal L4. L4 does not equal L5. L5 does not automatically equal L6. This distinction is central to the Week 6 integration assessment.*

## 5. Dependency Classification

| Code | Classification      | Application                                                                              |
|------|---------------------|------------------------------------------------------------------------------------------|
| SEQ  | Sequential          | A preceding output or condition is required before downstream progression                |
| TECH | Technical           | Compatibility between data, models, interfaces, components or environments is required   |
| INFO | Informational       | A downstream activity requires findings, specifications or other information             |
| ACC  | Acceptance          | Progression depends on the output satisfying a defined usability or acceptance condition |
| GOV  | Governance / Safety | Progression depends on approval, safety, knowledge, boundary or control conditions       |

*Multiple classifications may be assigned where a dependency has more than one control dimension.*

## 6. Master Integration & Dependency Register

| ID   | Provider            | Receiver                             | Required Input / Output                                              | Type        | Downstream Purpose                                      | Trigger                                           | Acceptance Condition                                               | Current State                | Mat.  | Downstream Exposure                                                        | PM Control / Next Action                                        |
|------|---------------------|--------------------------------------|----------------------------------------------------------------------|-------------|---------------------------------------------------------|---------------------------------------------------|--------------------------------------------------------------------|------------------------------|-------|----------------------------------------------------------------------------|-----------------------------------------------------------------|
| D-01 | Project Management  | All Workstreams                      | Approved baseline, scope and requirements                            | GOV         | Establishes authorised delivery parameters              | Throughout execution                              | Baseline remains approved, current and traceable                   | Controlled                   | L3    | Uncontrolled change or misalignment                                        | Maintain baseline governance                                    |
| D-02 | PM / Requirements   | All Workstreams                      | Validated problem definition and requirements                        | SEQ / GOV   | Provides common basis for solution development          | Before material solution decisions                | Requirements sufficiently defined for intended activity            | Controlled                   | L3    | Rework or inconsistent outputs                                             | Monitor material requirement changes                            |
| D-03 | Data Analytics      | Data Analytics                       | Appointment data and quality assessment                              | SEQ         | Establishes analytical foundation                       | Before dependent analysis                         | Data assessed for completeness, consistency and suitability        | Controlled                   | L3    | Analytical reliability may be constrained                                  | Maintain evidence of data-readiness assessment                  |
| D-04 | Data Analytics      | Data Analytics / Downstream Analysis | Analysis-ready dataset                                               | SEQ / ACC   | Enables attendance / no-show analysis                   | Before modelling and advanced analysis            | Required preparation and quality controls completed                | Controlled                   | L3    | Analytical delay or constrained findings                                   | Maintain dataset traceability and limitations                   |
| D-05 | Data Analytics      | Data Science                         | Validated attendance/no-show findings and relevant features          | SEQ / INFO  | Supports feature refinement and predictive development  | Before material model refinement                  | Findings supported, interpretable and demonstrably available to DS | Partially Evidenced / Active | L2–L3 | Predictive development may not fully reflect validated analytical evidence | Confirm downstream incorporation and resulting model change     |
| D-06 | Data Analytics      | ML / ML Engineering                  | Relevant analytical requirements and data considerations             | TECH / INFO | Informs processing and implementation interfaces        | During pipeline development                       | Requirements sufficiently clear and implementable                  | Active / Partially Evidenced | L2    | Interface or pipeline rework                                               | Confirm which analytical requirements were incorporated         |
| D-07 | Data Science        | ML / ML Engineering                  | Candidate model, feature requirements and modelling specifications   | TECH / SEQ  | Enables technical model integration                     | Before final model implementation                 | Candidate model and specifications reproducible and implementable  | Awaiting Input / Active      | L1    | Actual DS-model integration remains constrained                            | Obtain formal candidate-model handoff and receiver confirmation |
| D-08 | ML / ML Engineering | Data Science                         | Technical implementation and verification feedback                   | TECH / ACC  | Confirms implementation reflects modelling requirements | Following model integration                       | Expected model behaviour reproduced and verified                   | Conditional / Active         | L1–L2 | Potential divergence between model intent and implementation               | Complete verification against actual DS candidate               |
| D-09 | Data Analytics      | Generative AI                        | Relevant evidence-based support insights                             | INFO        | Informs support use cases where applicable              | When support design requires behavioural evidence | Insight demonstrably relevant to approved use case                 | Conditional / Not Activated  | L0    | Support design may proceed without relevant analytical evidence            | Activate only where a defined GenAI use case requires it        |
| D-10 | Generative AI       | Integrated Solution                  | Support capability, knowledge boundaries and escalation requirements | GOV / ACC   | Enables controlled patient-support capability           | Before integrated testing                         | Boundaries, grounding and escalation controls defined and testable | Active / Controlled          | L4–L5 | Safety, reliability or suitability exposure                                | Resolve identified disclaimer inconsistency and complete retest |
| D-11 | All                 | Project                              | Outputs, status and                                                  | INFO /      | Maintains                                               | Coordination                                      | Reporting is current,                                              | Active                       | L2–L3 | Inaccurate status or                                                       | Enforce evidence-based                                          |

| ID          | Provider            | Receiver                    | Required Input / Output                                      | Type       | Downstream Purpose                                        | Trigger                          | Acceptance Condition                                                       | Current State                                 | Mat.  | Downstream Exposure                                                  | PM Control / Next Action                                          |
|-------------|---------------------|-----------------------------|--------------------------------------------------------------|------------|-----------------------------------------------------------|----------------------------------|----------------------------------------------------------------------------|-----------------------------------------------|-------|----------------------------------------------------------------------|-------------------------------------------------------------------|
|             | Workstreams         | Management                  | dependency evidence                                          | GOV        | integrated project visibility                             | checkpoints                      | specific and evidence-backed                                               |                                               |       | delayed intervention                                                 | reporting                                                         |
| <b>D-12</b> | All Workstreams     | PM / Integrated Delivery    | Confirmed material cross-track handoffs                      | ACC        | Establishes that outputs have reached intended recipients | At each material handoff         | Receiver confirms receipt and usability                                    | Active                                        | L2–L3 | Apparent progress without downstream usability                       | Require explicit transfer and receiving-use evidence              |
| <b>D-13</b> | Data Science        | ML / ML Engineering         | Updated candidate model and implementation package           | TECH / SEQ | Enables actual technical model integration                | Model readiness point            | Required artefacts available, reproducible and accepted                    | Pending Workstream Evidence                   | L1    | ML pipeline cannot be confirmed against final candidate              | Confirm transfer, acceptance and model identity                   |
| <b>D-14</b> | ML / ML Engineering | Integrated Solution         | Technical component and interface evidence                   | TECH / ACC | Supports integrated workflow readiness                    | Integration-readiness assessment | Integration checks completed; no blocking technical defect                 | Substantially Evidenced / Integration Pending | L4    | Week 7 testing may be constrained if downstream use is not confirmed | Demonstrate actual incorporation into integrated workflow         |
| <b>D-15</b> | Generative AI       | Integrated Solution         | Refined assistant capability and safety/testing requirements | GOV / ACC  | Enables controlled AI-support testing                     | Before Week 7 testing            | Grounding, safety, escalation and boundary requirements testable           | Active / Conditionally Ready                  | L5–L6 | Incomplete testing or unresolved safety behaviour                    | Retest identified behaviour and confirm integrated use conditions |
| <b>D-16</b> | All Workstreams     | Week 7 Testing & Refinement | Testable outputs, known limitations and acceptance criteria  | SEQ / ACC  | Establishes controlled entry into Week 7 testing          | Week 7 entry point               | Required outputs available; limitations and acceptance criteria documented | Active / Readiness Preparation                | L4–L5 | Delayed or incomplete validation                                     | Track outstanding acceptance conditions through Week 7            |

## 7. Material Integration Interfaces

The register identifies four principal dependency chains requiring active PM attention. Each is shown below as a control flow rather than a simple description, so that Project Management can see, at a glance, where the chain is strong and where it is still unproven.

### 7.1 Analytics → Data Science



*Data readiness → validated attendance evidence → feature / model refinement*

Analytics has materially strengthened the evidence base through statistical validation, interaction analysis and revised interpretation of earlier findings.

The dependency is not yet fully closed, because evidence of exchange does not by itself establish that the receiving modelling process incorporated the relevant findings and that the incorporation produced a demonstrable downstream change.

*PM control: confirm the receiving-track use and resulting model-development change.*

### 7.2 Data Science → ML Engineering



*Candidate model → implementation → technical verification*

This is currently the most material technical dependency. ML Engineering has demonstrated substantial pipeline and scoring readiness, including schema alignment, batch scoring, integration tests, defect correction and reproducibility.

However, the actual Data Science candidate model has not been evidenced as formally handed over and accepted by ML.

*PM control: protect the handoff as a discrete integration event rather than treating ML pipeline readiness as completion of model integration.*

### 7.3 Analytics → Generative AI

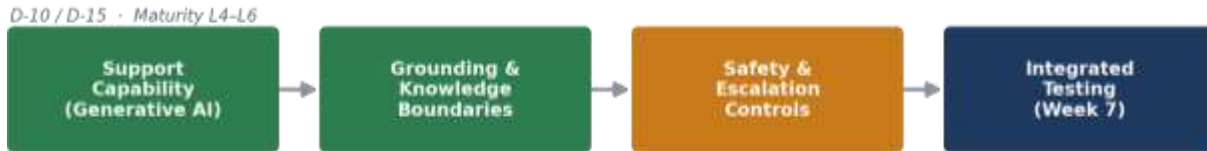


*Behavioural evidence → relevant support requirement → GenAI refinement*

This relationship is currently conditional. Analytics has produced evidence that could inform support design, but no material Analytics → GenAI integration has been established.

*PM control: activate and track this dependency only where an approved GenAI use case demonstrably requires the analytical output.*

## 7.4 Generative AI → Integrated Support



*Support capability → grounding / boundaries → safety / escalation → testing*

GenAI has progressed to scenario-based testing and identified a disclaimer-trigger inconsistency requiring refinement. This provides meaningful evidence for continued testing but does not, by itself, establish full solution-level validation.

*PM control: carry the identified behaviour into Week 7 retesting and confirm the conditions required for integrated use.*

## 8. Dependency Status Model

Dependency status must describe the condition of the dependency, not simply the amount of work completed.

| State                       | Definition                                                                                                  |
|-----------------------------|-------------------------------------------------------------------------------------------------------------|
| Controlled                  | Dependency is established and currently within acceptable control                                           |
| Active                      | Dependency is being worked or requires ongoing monitoring                                                   |
| Awaiting Input              | Required input has not yet been received                                                                    |
| Conditional                 | Dependency applies only where a defined condition or use case exists                                        |
| At Risk                     | Required condition or timing may not be achieved                                                            |
| Delayed                     | Required dependency has passed its expected point                                                           |
| Blocked                     | Downstream activity cannot meaningfully progress without resolution                                         |
| Resolved                    | Constraint has been addressed but may require confirmation before closure                                   |
| Closed                      | Dependency has been fully satisfied and requires no further control                                         |
| Pending Workstream Evidence | Relationship is established but Week 6 fulfilment is not adequately evidenced                               |
| Partially Evidenced         | Some elements are demonstrated, but transfer, acceptance, integration or downstream use remains unconfirmed |

## 9. Dependency Impact & Escalation Control

A dependency becomes a material PM concern when its failure could affect one or more of the following:

| Impact Dimension    | PM Assessment                                                                   |
|---------------------|---------------------------------------------------------------------------------|
| Schedule            | Does it alter sequencing, critical activity timing or milestone achievement?    |
| Quality             | Could downstream work proceed on an incomplete or unsuitable input?             |
| Integration         | Could incompatible or incomplete inputs create rework or interface failure?     |
| Readiness           | Could it prevent entry into Week 7 testing or subsequent implementation?        |
| Safety / Governance | Could proceeding without it compromise an approved control or safety condition? |

### Escalation levels

- Level 1 — Monitor: uncertainty or minor delay with no material downstream effect.
- Level 2 — Manage: dependency threatens an activity, handoff or acceptance condition and requires coordinated intervention.
- Level 3 — Escalate: dependency materially threatens a milestone, critical downstream activity, integration integrity, solution quality, safety/control condition or Week 7 readiness.

### Escalation trigger

*Material dependency exposure + downstream consequence + insufficient local resolution*

Where escalation is required, PM should create or update the relevant Issue, Risk, Decision or Schedule record rather than duplicating the matter here.

## 10. Dependency Ownership

### Provider

Accountable for producing the required input or output to the agreed condition.

### Receiver

Accountable for confirming whether the received output is fit for its intended downstream purpose.

### Project Management

Accountable for:

- Maintaining integrated dependency visibility;
- Monitoring material interfaces;
- Challenging unsupported dependency closure;
- Coordinating cross-workstream resolution;
- Assessing project and schedule exposure;
- Escalating material unresolved dependencies; and
- Maintaining linkage to other project controls.

### Project Leadership

Provides direction where dependency exposure exceeds the authority or resolution capacity of the workstream or PM function.

## 11. Review and Change Triggers

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The register shall be reviewed when any of the following occurs:

- A material dependency is created, removed or redefined;
- An expected output changes materially;
- A formal handoff occurs;
- a receiving workstream qualifies or rejects an output;
- An acceptance condition is not met;
- A dependency becomes At Risk, Delayed or Blocked;
- A dependency creates or materially changes a project risk;
- An integration issue emerges;
- A project decision changes the dependency relationship;
- Schedule sequencing is affected; or
- Week 7 testing readiness changes.

## 12. Week 6 Integration Control Position

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### OVERALL STATUS: AMBER — ACTIVE INTEGRATION CONTROL

Week 6 has moved the project beyond independent workstream execution into measurable interface control. The evidence demonstrates substantive advancement across all four technical workstreams; however, the maturity of the interfaces is uneven.

#### Current position

- Data Analytics — substantively ready. Analytical evidence has been strengthened through statistical testing, revised conclusions and deeper pattern analysis. Downstream predictive incorporation requires confirmation.
- Data Science — development active. The predictive approach, feature pipeline and candidate-model configurations have progressed, but executed comparative results and final candidate selection remain outstanding.
- ML Engineering — technically advanced. The scoring pipeline, schema alignment, integration tests and defect corrections provide strong technical readiness evidence. Integration against the actual Data Science candidate remains outstanding.
- Generative AI — Conditionally ready. Scenario testing has generated useful behavioural evidence and identified a refinement requirement. Broader reliability and integrated-solution validation remain open.

#### Principal dependency exposure

*Most material unresolved interface: Data Science candidate model → ML implementation and verification*

Other material control points are:

- Confirmation of Analytics → Data Science downstream use;

- Demonstration of ML → integrated workflow incorporation;
- Resolution and retesting of the identified GenAI behaviour;
- Confirmation of acceptance criteria for Week 7 testing.

#### PM determination

*The project has progressed from workstream-level development toward controlled integration, but full cross-workstream integration has not yet been demonstrated. Week 7 should therefore be entered as a controlled testing and refinement phase, with outstanding handoffs, acceptance conditions and integration evidence actively managed rather than treated as completed.*

## 13. Week 7 Dependency Horizon

The immediate dependency sequence — from Week 6 integration through to controlled entry into Week 7 testing — is set out below as a single control flow.

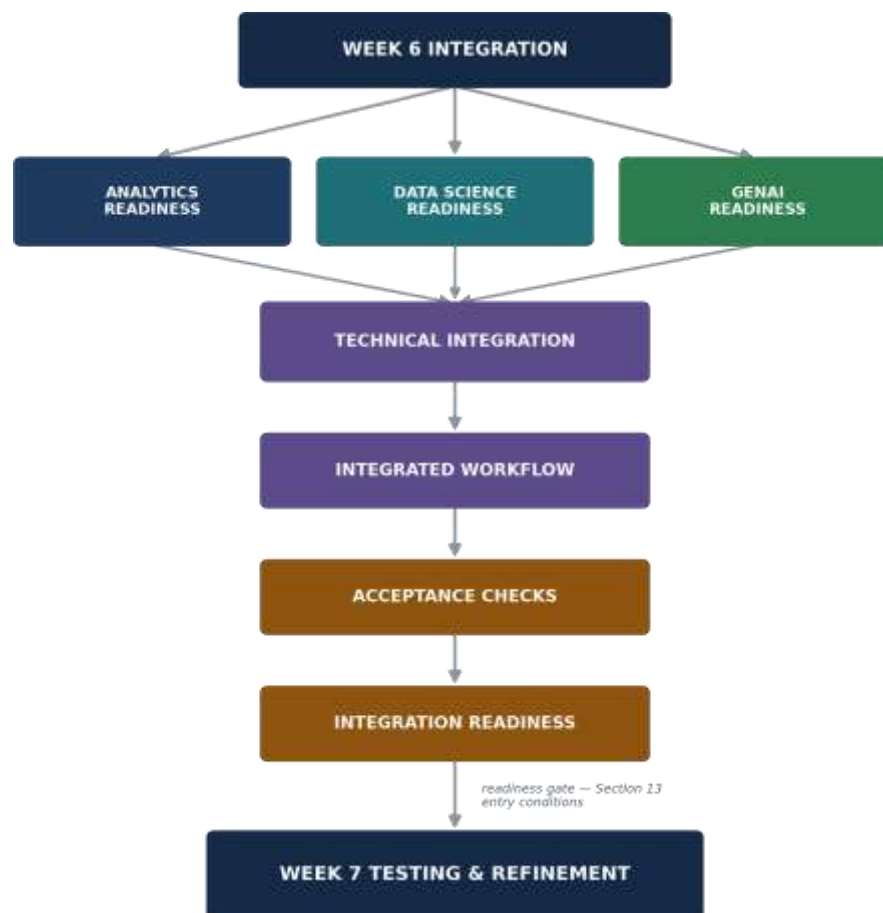


Figure 13.1 — Week 7 dependency horizon: readiness tracks converge through technical integration and acceptance before the Week 7 testing gate

#### Week 7 entry conditions

Before a material component is treated as ready for testing, PM should confirm:

- Required output has been transferred;

- Receiving workstream has confirmed usability;
- Downstream incorporation is identifiable;
- Known limitations are documented;
- Acceptance criteria are defined;
- Unresolved blocking issues are identified;
- Applicable safety/governance conditions are established; and
- Evidence is sufficient to support the planned test.

*This is a readiness control, not a requirement that every dependency be completely closed before testing begins.*

## 14. Register Governance

| Control              | Requirement                                                                                    |
|----------------------|------------------------------------------------------------------------------------------------|
| Register Owner       | Project Management Function                                                                    |
| Review Authority     | Project Sponsor / Project Leadership                                                           |
| Update Frequency     | At coordination checkpoints and following material dependency change                           |
| Status Basis         | Evidence-backed workstream reporting                                                           |
| Closure Basis        | Provider completion plus receiving-workstream confirmation where applicable                    |
| Change Control       | Material dependency changes assessed for downstream impact                                     |
| Cross-Reference      | Related Risk, Issue, Decision and Schedule records linked where applicable                     |
| Escalation Authority | Project Leadership for material unresolved exposure                                            |
| Review Trigger       | Dependency change, failed handoff, blocker, material risk, schedule effect or readiness impact |

## 15. Version Control

| Version | Period      | Purpose                                                                        |
|---------|-------------|--------------------------------------------------------------------------------|
| V1.0    | August 2026 | Initial dependency baseline                                                    |
| V2.0    | Week 5      | Execution-stage risk and dependency update                                     |
| V3.0    | Week 6      | Integration-focused dependency control and evidence-based readiness assessment |

***Next Review: Week 7 coordination review, or earlier following a material dependency change, failed handoff, blocker, schedule impact or integration issue.***